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How will our Values Fit Future Work? An Empirical Exploration of Basic Values and Susceptibility to Automation

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ABSTRACT
The discussion of how automation, especially intelligent technologies such as artificial intelligence (AI), affects the workforce is a focal point in scholarship about the future of work. In contrast to previous technological advances, routine and structured knowledge work are believed to be substituted with intelligent technological solutions. The paper responds to a recent call for research on how the implementation of intelligent technologies will affect the work-values dimension of person-job fit. The discussion of future work has hitherto focused on how the technologies change skill requirements at work. However, as routine and structured work is substituted, the values with which the work fits change as well. To explore how intelligent technologies can affect the work-values fit, the paper draws data from the European Social Survey and analyses the relationship between occupational values and automatability in 126 occupations. The results indicate that person-job fit can decrease in automatable occupations if intelligent technologies are implemented in the way scholars expect.

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automation; basic values; person-job fit; artificial intelligence; values

Introduction

Automation, especially artificial intelligence, has a focal role in the future of work debate. It is argued to affect society more profoundly than previous general-purpose technologies, such as electricity and the combustion engine, because it replaces cognitive tasks rather than enhances manual capabilities (McAfee and Brynjolfsson, 2016). The decline of middle-wage jobs since the 1980s in the US and Europe has been linked to the emergence of digital technologies that have decreased demand for routine-labour (Goos, Manning and Salomons, 2009, 2014; Autor and Dorn, 2013). Despite the negative impact the intelligent technologies have on the demand for routine-based jobs, the new technologies create new high-skilled jobs (Acemoglu and Restrepo, 2018b). However, they require different skill-sets than routine-intensive jobs which creates a discrepancy in skill demand and supply (Acemoglu and Restrepo, 2018a). Researchers largely agree that complementing AI relies on skills such as creativity and intuition (Brynjolfsson and Mcafee, 2012; Frey and Osborne, 2013); moral and critical judgement and adaptability (Hodgson, 2016); and perception, (physical) manipulation, and social intelligence (Frey and Osborne, 2013).

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These skills are difficult to code into algorithms and are therefore the human competitive advantage over machines. AI can complement the skills by freeing people from routine tasks to focus on using their creativity and other mainly human attributes (Brynjolfsson and Mcafee, 2012).

By replacing routine and structured work, the implementation of intelligent technologies affects the type of daily work required of workers. Such changes can lead to decreased person-job fit, and fit is widely acknowledged as a preferred state for both organisations and employees (Caldwell, 2011, 2017). Person-job fit affects, for example, job satisfaction and organisational commitment positively (Kristof-Brown, Zimmerman and Johnson, 2005). However, there are only a few studies that investigate how implementing technologies affects person-job fit and recently Halteh (*et al*., 2018) called for more research on how it could be affected by intelligent technologies. Technological solutions can create a person-environment misfit, which increases technologically induced stress (Ayyagari, Grover and Purvis, 2011). In a longitudinal study of sales force automation across two industries, researchers found that the implementation of automation decreased person-job fit over time (Speier and Venkatesh, 2002). This becomes especially important as the role of technologies at work is becoming more central. Some have referred to technology in working-life as becoming 'partners' rather than 'tools' (Bankins and Formosa, 2020). Indeed, scholars have argued that employees that can complement technologies will benefit most from intelligent technologies (Brynjolfsson and McAfee, 2014).

The discussion thus far has centred around which skills are required to complement intelligent technologies. Little is known about how the implementation of intelligent technologies affect the fit between values and work. However, a recent study has suggested that the transition to less structured and routine work decreases the fit between values and work (Långstedt and Manninen, 2020). According to the study, resistance to change occurs because change creates a 'misalignment' between values and work. That is, if the type of work required of workers changes as significantly as the researchers above suggest, the work-values fit decreases significantly following technologically driven change. A meta-analysis indicates that as the work-values fit decreases, work satisfaction and commitment decreases as well (Kristof-Brown, Zimmerman and Johnson, 2005). Considering the scope of the expected changes, the work-values fit is likely to pose challenges for maintaining the current levels of job commitment and satisfaction if the demand for routine and structured work decreases as estimated.

The aim of this study is to understand if different values are expressed in automatable and nonautomatable occupations and how strong the relationship between occupational values and occupational automatability is. Inferences of the potential effects of intelligent technologies for work-values fit are made based on the analysis. To the best of my knowledge, this is the first study to explore how intelligent technologies can affect work-values fit. The main argument is that if task replacement will occur as researchers have predicted, work-values fit can decrease because the remaining tasks cater to different 'needs, desires, or preferences' (Kristof-Brown, Zimmerman and Johnson, 2005) than the pre-automated work. The study is based on the European social survey and Frey and Osborne's (2013) automatability index. The data consists of over 32 000 European employees in 126 occupations.

Approaches to assessing the automatability of occupations

Whether the share of unemployment will increase, or not, as intelligent technologies are implemented during the next decades will be affected by far more than merely technological capabilities. The institutional environment in form of unions and other stakeholders affect how and what technologies are implemented and thereby limit the effects on employment (see Lloyd & Payne, 2019). Other barriers for implementing intelligent technologies are, for example, investment costs vs. labour costs, ethical and moral questions, and legislative and regulative restrictions. Questions arise such as, who is responsible for an accident where self-driving vehicles are involved (e.g. the owner or the manufacturer)? Or how long is the return on investment for automating low-wage jobs? All this, and more, affects the utilisation of intelligent technologies, and there is no certainty how they are resolved in the future.

Even within the technological capabilities perspective, conflicting assessments of how much work can be substituted with the novel technologies persist. Frey and Osborne's (2013) assessment of the automatability of occupations is probably most famous outside academia.

They assessed that 47% of the workforce in the USA is at 'high risk' for automation due to advances in sensor, robotics and digital technologies (Frey and Osborne, 2017). Their framework has been used in several different contexts illustrating different occupational structures across nations (see Bowles, 2014; Pajarinen and Rouvinen, 2014; Arntz et al., 2016 for country-specific assessments). In contrast, Melanie Arntz and her colleagues (2016) contend that 9% of employment in the USA and on average in the OECD countries can be automated. Adding to the diversity of assessments, a report by the McKinsey institute contends that less than 5% of occupations could be completely automated, however, intelligent technologies could substitute about 30% of tasks in 60% of occupations (McKinsey & Company, 2017). According to the report the most susceptible jobs relate to data collection, processing data, and predictable physical work.

There are many open questions about estimates of automatability and the variety of results they have produced bear witness to the difficulty of predicting changes in the labour market. The framework by Frey and Osborne, (2013) used in this paper has also been criticised. For example, the *routine task intensity index* (RTI) (Autor, Levy and Murnane, 2003) has been advocated as a more accurate measure of changes in employment in the USA during 2013–2018 than Frey and Osborne's assessment (Coelli and Borland, 2019). Whether similar results are obtained by using other assessments of automatability is beyond the scope of this paper but it is an interesting venue for developing this line of research. The framework has also been criticised for its treatment of occupations as task homogenous – tasks within occupations tend to vary significantly (Arntz, Gregory and Zierahn, 2016). This is, however, a critique that extends to any occupational level approach to work automation. Thus, it is crucial to complement occupational level research with individual level research. For this first exploration of how automation can affect work-values fit, the Frey and Osborne, (2013) framework is adopted. In the future comparing different automation frameworks' impact on work-values fit could provide valuable insights to the on-going debate on the future of work. More critical, however, is to study work-values fit on the micro-level and account for both intra-occupational value diversity and task heterogeneity.

Despite the large differences in the assessments, consensus is that automation mainly affects workers with low education and low-skills that perform routine tasks (e.g. Arntz et al., 2016; Brynjolfsson & Mitchell, 2017; Ford, 2013; Frey & Osborne, 2013; Hodgson, 2016). Therefore, social, non-regular physical, and creative tasks are expected to become more important in the future, because they are difficult to code into algorithms (Ford, 2013; Frey and Osborne, 2013; Arntz, Gregory and Zierahn, 2016; Hodgson, 2016; McKinsey & Company *et al*., 2017). These assessments do not necessarily mean that jobs vanish, but they imply that the *type of work* in jobs change.

The relationship between values and automation

The first step to infer how intelligent technologies can affect the work-values fit is to understand how values relate to the characteristics ascribed to future of work and which values are relevant for those characteristics. For this purpose, prior research is reviewed to establish which values are relevant to the future of work debate and how. The next step is to analyse the relationship between values and occupational automatability. These steps provide a basis to infer how the implementation of intelligent technologies impacts the work-values fit.

A central link between automation and basic values is that people with similar value priorities are attracted to similar *types* of occupations (Holland, 1985; Gandal *et al*., 2005; Arieli, Sagiv and Cohen-Shalem, 2016). Recent studies find that openness-to-change values correlate positively with level of education while conservation values correlate negatively with level of education (e.g. Schwartz, 2006). A line of research in the seventies argued that people in different *types* of occupations possessed different values (Hitlin and Piliavin, 2004): Employees in routinised jobs valued conformity while employees in jobs with less supervision valued selfdirection. Recent studies of vocational choices and occupational values confirm these findings (Sagiv, 2002; Knafo and Sagiv, 2004; Arieli, Sagiv and Cohen-Shalem, 2016). People performing different *types* of work tend to prioritise different values.

The theory of basic human values

Basic values are beliefs about what is desirable and what means are acceptable in the pursuit of those desires (Rokeach, 1973; Schwartz, 1992). Basic values are central to how we evaluate our environment and other people, and how we behave (Bardi and Schwartz, 2003; Cieciuch, 2017; Schwartz, 2017). They are based in biological, social, and psychological needs that vary in their relative importance (Schwartz, 1992) (see Table 1 for definitions). The value structure, depicted as a circle of ten values arranged according to their compatibility, represents the relationship between different values (Schwartz, 1992; Schwartz *et al*., 2012; Borg, Bardi and Schwartz, 2017). Adjacent values are compatible while values opposing each other are incompatible (see Figure 1). The variance in the relative importance of the value types illustrates different value priorities; this is called a value hierarchy (Schwartz, 1992). Therefore, despite the universal structure, the value priorities, or value hierarchies, vary across different samples, but maintain the circular form in Figure 1. The value hierarchy is similar across the world (Schwartz and Bardi, 2001). Comparing people that are socialised into different institutional environments, such as

Table 1. The defining goals of basic human values from Schwartz (1992).

Upper-level value	Value type	Defining goal	Upper-level Value	Value type	Defining goal
	Benevolence	Focus on the welfare of people with whom one is in close contact		Tradition	Respect, commitment, and acceptance of the customs and ideas that one's culture or religion impose on the individual.
Self-transcendence	Universalism	Understanding, appreciation, tolerance, and protection for the welfare of *all* people and for nature	Conservation	Conformity	Restraint of actions, inclinations, and impulses likely to upset or harm others and violate social expectations and norms.
				Security	Safety, harmony, and stability of society, of relationships, and of self.
	Self-direction	Independent thought and action - choosing, creating, exploring.		Power	Attainment of social status and prestige, and control or dominance over people and resources
Openness-to change	Stimulation	Excitement, novelty, and challenge in enhancement life.	Self-enhancement	Achievement	Personal success through demonstrating competence according to social standards.
	Hedonism	Pleasure or sensuous gratification of oneself.			

occupations, has been argued to be more fruitful than national comparisons (Fischer and Schwartz, 2011).

The value types relate directly to work values, they are a broader set of values that work values specify (Schwartz, 1999) for example, openness-to-change values correlate with the work values prestige and intrinsic because they relate to personal growth and autonomy (Ros, Schwartz and Surkiss, 1999). Recent reviews of research on basic values

Figure 1. Schwartz (1992) value structure.

and work related behaviour clearly show that basic values are relevant for many aspects of working life (for thorough reviews see Sagiv, Schwartz and Arieli, 2011; Arieli and Tenne-Gazit, 2017; Arieli,Sagiv and Roccas, 2019).

Values and skills of the future

The skills that are expected to be in demand in the future are often presented in terms of upskilling the workforce. The argument rests on the premises that as jobs become augmented by automating structured tasks, what is left are tasks that are difficult to code into algorithms (e.g. Ford, 2013; Frey and Osborne, 2013; Manyika *et al*., 2017). This might, however, not be the case. Social robots can, for example, replace labour in some relatively high skilled and social tasks (Bankins and Formosa, 2020), such as in healthcare. Another stream of research argues that intelligent technologies can lead to down skilling. Technologies are argued to deprive humans of the opportunity to develop and utilise skills (Vallor, 2015). Consider the learning process required for performing data analysis and data collection, which are expected to be automatable tasks due to their structured nature, and how replacing those tasks could erode the skill of creating overviews and understanding conceptual and empirical relationships. According to the studies reviewed above, the main task that would remain for humans would be convincing other humans, which certainly shifts the *type* of work required.

Creativity is frequently reported as a central skill in future work (e.g. Ford, 2013; Frey & Osborne, 2013; Mitchell & Brynjolfsson, 2017). Creativity has been perceived as a trait, a process, and based on certain values (Dollinger, Burke and Gump, 2007). A central aspect of creativity is the ability to create something that diverges from conventional thinking (Sternberg, 2010). The ability to be creative is influenced by both individual and social elements that support the creation of original things (Amabile, 1983; Amabile *et al*., 2004; Sternberg, 2010; Acar, Burnett and Cabra, 2017). The individual elements involve, for example, skills, knowledge, thinking styles, personality, motivation, and decision-making. The social elements consist of leadership, management practice, peer support, expectations, and job characteristics (Amabile, 1983; Amabile *et al*., 2004; Sternberg, 2010).

Previous research indicates that creativity is positively linked to self-direction, stimulation, and universalism values (Dollinger, Burke and Gump, 2007; Kasof *et al*., 2007; Lebedeva *et al*., 2019). Table 2 provides an overview of the research to date on values and the skills identified as relevant in the future of work literature. Self-direction

Table 2. Values that are positively associated with the skill requirements of the automated future.

Skills	Values	References
Creativity	Self-direction, universalism, stimulation.	Dollinger *et al*., 2004; Kasof *et al*., 2004; Lebedeva *et al*.2019
(Pro)social behaviour	Universalism, benevolence, self-direction, achievement	Schwartz, 2010; Sagiv & Schwartz, 1995; Schwartz, 2007
Proactive behaviour (e.g. organisational change, suggestion-making)	Openness-to-change	Seppälä *et al*., 2012
Change attitudes to voluntary change imposed change	Openness-to-change (+) Conservation (-) Conservation (+); Openness-to-change (-)	Oreg *et al*., 2008; Sverdlik & Oreg, 2009; 2015

comprises the motivational goal to define one's thoughts and actions independently (Schwartz, 1992). This is a central aspect of diverging from conventional thinking. Stimulation is linked to creativity through the motivation to seek new opportunities and excitement (Kasof *et al*., 2007), and universalism supports creativity by comprising values such as tolerance and world of beauty (Dollinger, Burke and Gump, 2007). These values are important in relation to thinking styles and knowledge because they comprise a motivation to investigate and develop ideas and expand horizons. In contrast, security, conformity, and power values have a converse relationship with creativity (Dollinger, Burke and Gump, 2007; Kasof *et al*., 2007). Creativity involves taking risks (Shalley and Gilson, 2004); therefore, security as a motivation opposes creativity. Similarly, conformity and tradition opposes creativity as they confine thinking to group norms, which contradicts creativity as divergent thinking (Sternberg, 2010). Thus, self-direction, stimulation, and universalism are related to the creative skills required in the future (see Table 2).

Social skills have been pronounced important skills for the future as well (e.g. Frey & Osborne, 2013). They are important in, for example, management, contract negotiation, and innovation; it is important to persuade people about ideas, services, and products. Caregiving and teaching requires these skills, which is reflected in the prevalent values in the professions. For example researchers have found that self-transcendence values are more valued by social workers than other professions (Knafo and Sagiv, 2004; Tartakovsky and Walsh, 2018). In a study of values and empathy, Silfver *et al*., (2008) found that empathetic concern and perspective-taking was associated with those same values. Further, the benevolence and universalism values are related to collaborative behaviour (Arieli, Sagiv and Roccas, 2019). These associations are explained by the motives that the value types represent. Benevolence values strive to ensure the welfare of close others and the goal of universalism is to care for the welfare of everyone. Social skills require the ability to consider other people's experiences. The benevolence and universalism values express the importance of other people's experiences. Therefore, these values are associated with the social skills associated with work in the future.

Changing work environments

A work environment is the 'situation or atmosphere created by the people who dominate a given environment' (Holland, 1973, 27). When personality types are matched to their corresponding work environments it results in work satisfaction, achievement, and vocational stability because the employee's needs are met and supported by the environment (Holland, 1973). A central aspect of the relationship between values and work environments is that the work environment affects which values are relevant during the selection of occupations and what values are reaffirmed within the occupation (Hitlin and Piliavin, 2004). This is reflected in the relationship between values and work environments (Knafo and Sagiv, 2004) and occupational choices (Sagiv, 2002; Arieli, Sagiv and Cohen-Shalem, 2016). If people work in environments that do not match their personality type, they are confronted with requirements of competencies they do not possess, worldviews they do not hold, depreciation of their values, and tasks that are in stark contrast to their preferences (Holland, 1985). The central tenet in Holland's theory is that people have different needs and competencies and seek work environments where they can realise them. The work environment typology involves six environment types corresponding to

Table 3. Definitions of work environments cited from Holland (1985) and their relationships to values (+ means positive association – means negative association).

Work Environment	Defining characteristic of tasks in the work environment	Correlated values of employees in different environments (Knafo and Sagiv, 2004)	Correlated values with vocational choice of environment (Sagiv, 2002
Realistic	The explicit, ordered, or systematic manipulation of objects, tools, machines, and animals.	Hedonism (+), Tradition (+), Benevolence (-), Self-Direction (-)	No significant correlations
Conventional	The explicit, ordered, systematic manipulation of data, such as keeping records, filing materials, reproducing materials, organizing written and numerical data according to presribed plan, operating business and data processing machines.	Power (-), Security (+)	Security (+), Conformity (+), Tradition (+), Universalism (-), Self-direction (-), Stimulation (-)
Investigative	The observation and symbolic, systematic, creative investigation of physical, biological, or cultural phenomena.	Hedonism (-), Tradition (-), Self-direction (+)	Universalism (+)
Artistic	Ambiguous, free, unsystemized, activities and competencies to create art forms or products.	Achievement (+), Conformity (-)	Security (-), Conformity (-), Universalism (+), Self-direction (+)
Social	Manipulation of others to inform, train, develop, cure, or enlighten.	Achievement (-), Benevolence (+), Universalism (+)	Benevolence (+)
Enterprising	The manipulaiton of others to attain organizational or self-interest goals.	Universalism (-), Achievement (+), Power (+)	Achievement (+), Power (+), Universalism (-)

six personality types: realistic, investigative, artistic, social, enterprising, and conventional (see Table 3 for definitions).

Conventional occupations are of particular interest because they involve jobs that are associated with middle-wage income and structured work. These types of occupations involve central aspects of what Frey and Osborne, (2013) identify as automatable tasks like organising data in standard ways. This type of work has seen a steady decline in the last decades (Goos, Manning and Salomons, 2009). It is unlikely that the new work environment after automation would correspond to the same values as the prior work environment if it changes as anticipated, which implies a shift from conventional to investigative or artistic work environments. Knafo and Sagiv's, (2004) and Sagiv's, (2002) research indicates that the latter and former occupations are characterised by values at opposing ends of the value structure – which indicates a forthcoming work-values misalignment. Such misalignment is likely to occur if the current trend of automating routine tasks continues and demand for creative and social work increases. The high-level of susceptibility to automation exhibited by these types of occupations is related to the predominance of standardised cognitive tasks and the lack of ambiguous and creative tasks. The transition to complementing intelligent technologies may be difficult in these occupations because values that relate negatively to creativity are prioritised within them.

Realist occupations share the characteristics of Frey and Osborne's (2013) medium-risk category: both involve the manipulation of irregular objects that occurs in, for example, maintenance and mechanics work. These occupations have a positive association with tradition and hedonism and a negative association with self-direction and benevolence (Knafo and Sagiv, 2004). The occupations are not as susceptible to automation as conventional occupations because they involve adaptation to different situations and

the ability to work in cramped spaces, for example performing repairs in different facilities. The lower level of susceptibility to automation exhibited by these occupations is related to overcoming engineering bottlenecks in robotics rather than codifying advanced cognitive tasks. Once the bottlenecks are solved, previous research would suggest that realist occupations are faced with similar challenges as conventional occupations.

To summarise, the relationship between values and automation rests on two fundaments. First, people seek work that aligns with the pursuit of their values (Ros, Schwartz and Surkiss, 1999). Research on values in occupations (e.g. Knafo & Sagiv, 2004) gives reason to expect that automation may create a misalignment of values and the work available in the future because the values associated with, for instance, the conventional occupations differ from those in investigative, implying that the occupations support different values. The depreciation of the tradition value in investigative work does not correspond with the appreciation of tradition in conventional jobs, which creates a misalignment between values and the nature of work (Table 3). Second, values are associated with several skills that are expected to become increasingly demanded as automation progresses. Particularly skills related to creativity and social skills are of interest. From the review above (Table 2) we find that these skills are associated with values such as self-direction and universalism. Both value types are less prevalent in the conventional and realist occupations that resemble the automatable occupations in Frey and Osborne's (2017) framework. Thus, in addition to the compatibility of the work environment, the changing skill requirements following automation makes values relevant. It can be expected that benevolence, universalism, self-direction, stimulation, and achievement are negatively associated with automatability, while tradition, conformity, and security are positively associated with automatability. Power and Hedonism are ambiguous, for example power relates negatively to general creativity (Kasof *et al*., 2007) and positively to innovation at work (Purc and Lagun, 2019). Power is further positively associate with enterprising work such as management (Sagiv, 2002; Knafo and Sagiv, 2004; Gandal *et al*., 2005). Thus the following hypotheses are proposed:

H1: Benevolence, universalism, self-direction, stimulation, and achievement correlate negatively with automation and are more prevalent in non-automatable occupations.

H2: Tradition, conformity, and security correlate positively with automation and are more prevalent in automatable occupations.

Data and Method

The data is derived from the European social survey (ESS) round 8 (European Social Survey, 2016). The 8th round included 44,387 respondents comprising representative samples from 23 European countries. The data is evenly distributed between the countries. The highest share of responses for a country is 6.7% (Germany) and the lowest share is 2.3% (Iceland) with a mean of 4.4% (see appendix A). The sampled countries are representative of 92% of the EU population in 2016 (based on Eurostat), with the addition of representative samples from the Russian Federation and Israel. The survey includes

a 21-item version of the Schwartz portrait values questionnaire (PVQ), which measures 10 value types in Figure 1.

The PVQ asks "tell me how much each person is or is not like you?' followed by a description that corresponds to a personal value. The responses are measured on a scale that ranges from 1–6 where 1 = "very much like me,' 2 = "like me,' 3 = "moderately like me,' 4 = "a little like me,' 5 = "not like me,' and 6 = "not like me at all.' The values were recoded into the opposite values for the interpretation of the dataset so that 1 = 6, 2 = 5 and so forth. The PVQ questionnaire requires that each respondent's scores are centred to avoid scale use bias, which is caused by preferences either to attribute scores in surveys to the extremes or the middle of the scale. This is achieved by subtracting the mean rating of each individual from its responses (Schwartz, 2009). This was performed by using the SPSS syntax that ESS provides on their web page. As Schwartz (Schwartz, 2009, 2012) instructs, missing values are dropped.

Utilising the Frey and Osborne, (2013) categorisation with the ESS data required translating the international standard classification of occupations 2008 (ISCO-08) to the American Standard Occupational Classification (SOC). A list from the US Bureau of Labour Statistics was used to match Frey and Osborne's (2013) assessment of SOC occupations with the ISCO-08 categorised ESS data. The occupations were assigned a level of susceptibility for automation and recoded into a "risk level' variable with values 1 = low risk (automatability .0-.3), 2 = medium risk (.301-.7), and 3 = high risk (.701–1). The characterisation of occupations is based on the 2010 O*NET service maintained by the US Department of Labour. The characteristics are assisting and caring for others, persuasion, negotiation, social perceptiveness, fine arts, originality, manual dexterity, finger dexterity, and cramped work space. These characteristics represent engineering bottlenecks that decrease the automatability of an occupation. The risk groups are demographically alike, they differ from each other mainly in terms of education level and a larger share of men in the high-risk groups while the medium- and low-risk groups have a majority of women. The mean educational level is higher in the medium- and low-risk group than in the high-risk group, as could be expected based on previous research (e.g. Arntz et al., 2016) (see (see Table 4 for definitions) for definitions).

Occupations with less than 99 respondents were merged with their parent ISCO-08 category. For example the occupations 1111, 1112, 1113, and 1114 were coded into the category 1100. The susceptibility of the parent category is the mean of the parent and subcategories (e.g 1100 is the mean of 1100, 1111, 1112, 1113, and 1114). If the re-categorisation would not amount in a category over 99 responses, the category was excluded from the analysis. This led to a total of 126 occupational categories with a susceptibility score. 37 occupations were categorised as low-risk, 39 as medium risk, and 50 as high-risk. These categories comprise 32,324 individual respondents. Following the categorisation of the occupations, the means of each occupation were calculated to represent the value tendencies within the occupations (cf. the study of Knafo& Sagiv, 2004) By doing so, the level of analysis corresponds to the "level of theory' (i.e. occupational level values and occupational level automatability) (Klein, Dansereau and Hall, 1994). Correlations between automatability and values on the occupational level were performed to analyse the relationship. Age and education level were controlled for because previous research indicates their importance in relation to values, while the relationship between values and sex and nationality has received contradictory results

Table 4. Demographics of the risk groups. Education level 3: lower tier upper secondary, level 4 upper tier secondary, level 5 advanced vocational.

Descriptive statistics			Gender		Level of education			Age	
			Male	Female	Total	Mean	Std. Dev.	Mean	Std. Dev.
Risk level	Low-risk occupations	Count	4188	5344	9532	5,46	2,715	50,45	16,565
		% within Risk level	43,9%	56,1%	100,0%				
	Medium-risk occupations	Count	4750	5745	10,495	4	3,041	50,33	17,528
		% within Risk level	45,3%	54,7%	100,0%				
	High-risk occupations	Count	6775	5732	12,507	3,49	2,62	50,58	18,255
		% within Risk level	54,2%	45,8%	100,0%				
Total		Count	15,713	16,821	32,534	4,23	2,907	50,46	17,539
		% within Risk level	48,3%	51,7%	100,0%				

(Schwartz and Rubel, 2005; Schwartz, 2006; Fischer and Schwartz, 2011) and the level of education is expected to have a negative impact on the automatability (Arntz, Gregory and Zierahn, 2016; Frey and Osborne, 2017). To test whether occupations in the different risk groups differed an ANOVA was performed. The normal distribution of the data was visually analysed through histograms and p-plots. No outliers were found. The effect size (*r*) of the ANOVA was calculated using the formula $r = \sqrt{\frac{ssm}{sst}}$ (Field, 2013).

Results

The correlations indicate a statistically significant medium effect size for most values, with the exception of hedonism and achievement that show a non-significant relationship to the automatability of occupations. The correlations follow the theorised pattern: Security ($r = .61$ $p < .00$), tradition ($r = .44$ $p < .001$), conformity ($r = .43$ $p < .001$), power ($r = .24$ $p < .008$) correlate positively with automatability, while self-direction ($r = -.65$ $p < .001$), stimulation ($r = -.22$ $p = .014$), universalism ($r = -.42$ $p < .001$), and benevolence ($r = -.33$ $p < .001$) correlate negatively with automation. The *r* scores when controlling for mean age increases the correlation between automatability and stimulation ($r = -.25$ $p = .004$), tradition ($r = .51$ $p < .001$), and conformity ($r = .5$ $p < .001$). When controlling for mean level of education only selfdirection ($r = -.19$ $p = .036$) and power ($r = .2$ $p = .024$) maintain their statistical significance, the correlation with the other values become statistically non-significant. The direction of the correlations are maintained except for tradition that becomes negative and hedonism that becomes positive and increases effect size (see Figure 2). Importantly the correlations in Figure 2 follow a sinusoid shape which indicates a systematic relationship between values and the external variable (i.e. automatability) (cf. Schwartz, 1992, 1996) (see Table 5 for definitions).

To analyse if the correlations manifest as differences between the risk groups defined by Frey and Osborne, (2017) an analysis of variance (ANOVA) was performed using Scheffe and Games and Howell's post hoc tests. The tests indicate that high-risk occupations differ

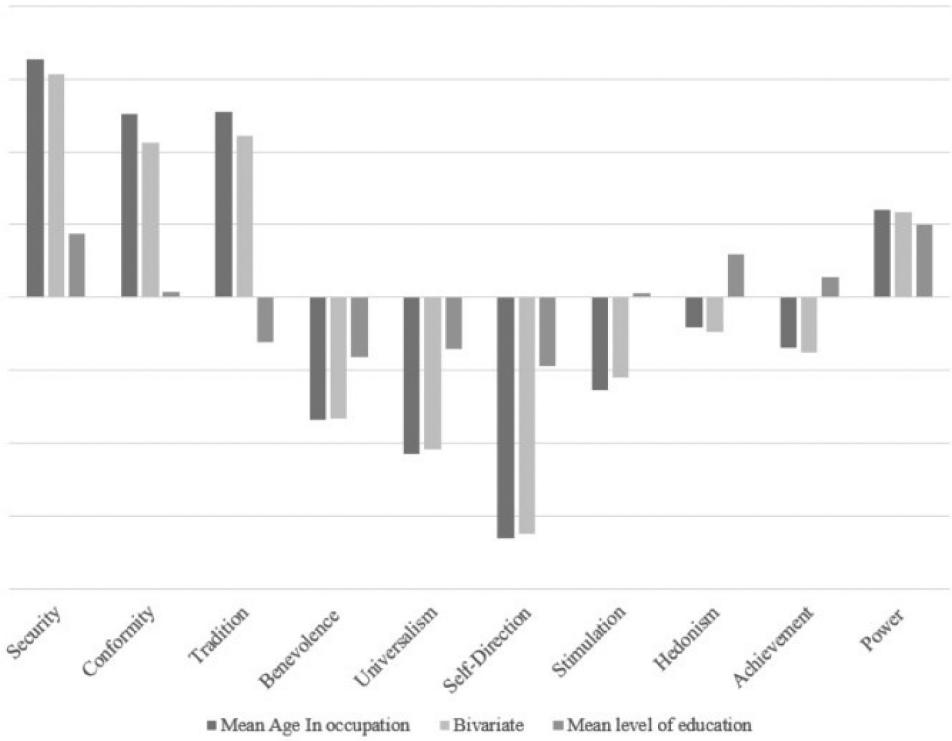

Figure 2. Bivariate and partial Correlations between values and automatability controlling for age and highest level of education.

significantly from medium- and low-risk occupations in all values except achievement, hedonism, and stimulation (see figure 3 for group means). The test reported no significant differences between the risk groups in these values (r = .08-.0.17 p > .05). For the values security and self-direction, all groups differed significantly from each other (p < .05). The low- and medium-risk groups did not differ significantly in universalism (p = .06) and benevolence (p = .71) while the high-risk group differed significantly from both (p < 0.05). This indicates that the medium risk and low-risk groups share values related to the welfare of in-group members, but not an as strong concern for the welfare of people beyond it. The low-risk group differed from the medium- and highrisk groups in the conformity and tradition values, however, the medium- and high-risk groups did not differ significantly in these values (con p = .44, tra p = .15). The low- and medium-risk groups differ significantly from the high-risk group regarding the power value (p = < .05). The ANOVA table with calculated effect sizes is available in appendix C and mean scores in appendix D.

The results supported both hypotheses, however, education rose as an important mediator of the value-automatability relationship. This finding is in line with both the research on the automatability of occupations and values research. Even conflicting studies within automatability research have found that higher levels of education is a factor that decreases susceptibility to automation (Frey and Osborne, 2013; Arntz, Gregory and Zierahn, 2016). In values research education has been consistently linked to openness-to-change values (e.g. Hitlin and Piliavin, 2004; Schwartz, 2006), thus it could be expected that education mediates the relationship between values and automatability.

Table 5. Bivariate and partial correlations between values and automatability. Control variables mean age in occupation and mean education level in the occupations.

Control Variable		Security	Conformity	Tradition	Benevolence	Universalism	Self-Direction	Stimulation	Hedonism	Achievement	Power
Mean Age	Correlation	.654	.502	.509	−.336	−.431	−.663	−.254	−.081	−.140	.240
	Significance (2-tailed)	.000	.000	.000	.000	.000	.000	.004	.369	.120	.007
	df	123	123	123	123	123	123	123	123	123	123
Mean level of education	Correlation	.173	.015	−.123	−.164	−.142	−.188	.011	.118	.056	.201
	Significance (2-tailed)	.054	.864	.171	.068	.115	.036	.905	.189	.537	.024
	df	123	123	123	123	123	123	123	123	123	123
Bivariate	Pearson	.613	.426	.444	−.332	−.418	−.650	−.219	−.096	−.150	.235
	Correlation	.000	.000	.000	.000	.000	.000	.014	.285	.094	.008
	Sig. (2-tailed)										
	N	126	126	126	126	126	126	126	126	126	126

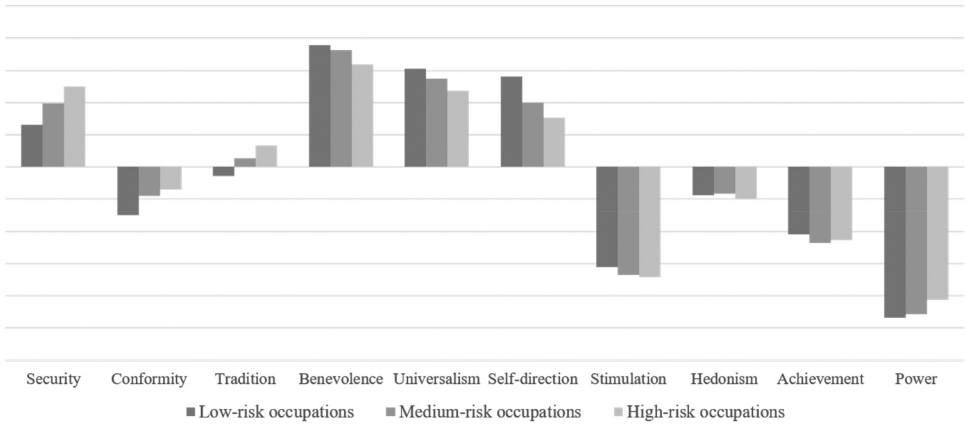

Figure 3. Mean scores of values in occupations at different level of risk for automation.

Discussion

This study answers a recent call to study how the implementation of intelligent technologies can affect the person-environment fit (Halteh *et al*., 2018) and more specifically, a recent call for research into the work-values fit in relation to changes following the implementation of intelligent technologies (Långstedt and Manninen, 2020). The observations indicate that the substitution of work with intelligent technologies can lead to a decreased work-values fit. Like change in general (Caldwell, 2017), implementing intelligent technologies creates a shift in the tasks that employees need to perform. In particular, it replaces routine-like work and employees are left with creative or social work. This can disrupt the fit between values and work (Långstedt and Manninen, 2020). Researchers have established that within particular occupations the implementation of technologies decreases person-job fit (Speier and Venkatesh, 2002; Ayyagari, Grover and Purvis, 2011). This paper contributes to the person-job fit and future of work literature by presenting a broader perspective on how the implementation of intelligent technologies could affect work-values fit at the occupational level. The study reveals a potential challenge that has thus far been overlooked by researchers.

The relationship between automatability and basic values is linked to the values that work satisfies and the skills jobs require. According to Holland, (1985) people seek work environments that correspond to their values and skills. Holland's theory has been corroborated by more recent research into values and work (Knafo and Sagiv, 2004; Gandal *et al*., 2005; Arieli, Sagiv and Cohen-Shalem, 2016). Work acts as an important venue for pursuing values and therefore similar types of work attracts people that share similar value priorities (Ros, Schwartz and Surkiss, 1999; Sagiv, 2002; Hitlin and Piliavin, 2004). Automation affects specific types of work (i.e. structured and routine-like) and therefore automatable occupations share similar value tendencies. Thus, the relationship observed in this study between basic values and automation emerges. This is not to claim that there is a causal link between values and automation. That is, automation does not affect values and it is unlikely that values affect the automatability of occupations. As a suggestion for future research, one might, however, entertain the idea that automatable occupations have lagged behind in automating routine tasks because of their relative prevalence of conservation values in comparison to non-automatable occupations.

Conservation values share a motivation to maintain a status quo and avoid uncertainty (Schwartz, 1992) and therefore occupations that are characterised by conservation values could be less inclined to renew their ways of working in comparison to occupations where openness-to-change values prevail.

That the relationship between values and automation is strongly mediated by the level of education is supported by other studies showing that both the level of education and the subject of education is closely linked to values (e.g. Schwartz, 2006; Arieli, Sagiv and Cohen-Shalem, 2016). This has been linked to both attraction and socialisation (Hitlin and Piliavin, 2004; Arieli, Sagiv and Cohen-Shalem, 2016). Educational programs vary in which values they appeal to and within those programs certain values are encouraged and therefore their importance for students within the programs increases (Schwartz, 2006). Value differences are already visible at early educational phases when vocational choices are made (Sagiv, 2002), which is corroborated by the differences in value priorities between university students of different subjects (Arieli, Sagiv and Cohen-Shalem, 2016). Thus, the choice of occupation and education seems intimately related to values and as a result occupations that require different educational backgrounds differ in their value priorities (Hitlin and Piliavin, 2004; Knafo and Sagiv, 2004). Further, the level of education is a decisive factor that creates resilience to automation (Arntz, Gregory and Zierahn, 2016; Frey and Osborne, 2017). As a result, education becomes an important link in the relationship between occupational values and automatability.

The results of the analysis indicate that there is a strong relationship between automatability and the majority of Schwartz' (1992) value types. Previous research provided background to which of the values would be prevalent within the different risk groups. The study shows clear indications that self-direction, stimulation, universalism, and benevolence values are associated with occupations resilient to automation. In contrast, power, conformity, and tradition values are linked to higher automatability. This indicates that if work becomes less structured, the person-job fit may decrease in automatable occupations as suggested by previous studies (Speier and Venkatesh, 2002; Ayyagari, Grover and Purvis, 2011). This can become a significant issue because the alignment of values with work correlates positively with job satisfaction and organisational commitment (Kristof-Brown, Zimmerman and Johnson, 2005; Ostroff, Shin and Kinicki, 2005; Edwards and Cable, 2009). Thus, the misfit following the implementation of intelligent technologies can have a detrimental impact on working life. Whether a significant share of the workforce will be replaced or not, the value tendencies of replaceable occupations put them at a disadvantage if structured work is replaced. If work becomes more social and creative by nature, the replaced workers are likely to encounter difficulties to find jobs where they can pursue conformity, tradition, security and power because these values are attained through structured tasks. This makes the implications of the study relevant for any organisation that strives to achieve the benefits of a high level of work-values fit.

Practical implications

The results indicate that person-job fit can decrease in many occupations as a consequence of implementing intelligent technologies. It is paramount that organisations do not limit their efforts to the skills-work dimension of person-job fit but integrate the work-values fit as well if the changes occur as estimated. Organizations need to consider how changes affect

the fit between values and work requirements. If work becomes more dynamic as researchers suggest, it is important to acknowledge that this can conflict with the prevalent values and create inertia in the change process (Långstedt and Manninen, 2020). Furthermore, decreases in work-values fit are reflected in decreases in job satisfaction and commitment (Kristof-Brown et al., 2005). Thus, the organisations implementing such changes should alleviate the impact the change has for work-values fit. This could be achieved by, for example, making the work environment more stable and predictable.

The central role education has as a mediator of the relationship between values and automatability suggests that policy-makers should make life-long learning a priority to ensure a competent and relevant labour force as previous researchers have suggested (e.g. Ghislieri, Molino and Cortese, 2018). However, given that the results indicate that work-values fit may become an issue, interventions in early childhood can be worth pursuing. This could involve making childhood education less streamlined and more dynamic to, for example, provide an environment where children have the opportunity to be creative and express themselves (Haslip and Gullo, 2018). However, the current trend, has been for some time to produce streamlined education that rewards conformity rather than creativity and exploration (Prentice, 2000; Haslip and Gullo, 2018).

Future directions and limitations

There are some major trajectories that future research should take based on the results and discussion of the paper. The first, and perhaps most practitioner-oriented, trajectory is to study how management practices can alleviate potential work-values misfit and what the broader institutional field's role is in alleviating it. The second trajectory is to investigate how the relationship between values and job automatability emerges on the individual level. Such study would help us understand further the relationship between types of jobs and values and more specifically how values relate to specific types of automatable tasks. The third trajectory is to understand how the relationship between values and automation has evolved during the past decades. Have, for example, values in susceptible occupations shifted towards openness-to-change as automation within the occupation has progressed? These are central questions that would provide insights into how organisations and nation states can adapt to the potential change in working life following the potential diffusion of intelligent technologies.

Analysing automatability at the occupational level has its weaknesses. Critical voices have questioned whether it is reasonable to assume that occupations are homogenous enough to be treated as the unit of analysis. The main argument is that occupations involve a plurality of tasks and jobs within occupations are therefore prone to vary (Arntz, Gregory and Zierahn, 2016).

Furthermore, Frey and Osborne's framework has received critique for its methodology and has been a poorer predictor of automation than the RTI framework (Coelli and Borland, 2019). Central to the paper is that technology can perform certain types of tasks and thus changes the work environment. Similarly, values tend to relate more to individuals than to social groups (Fischer and Schwartz, 2011). The use of occupation as the level of measurement is, however, supported by Schwartz and Bardi, (2001), who argue that occupations may be consistent with "cultural groups', that share socialisation into certain values and institutional environments.

Despite considerable methodological challenges with cross-level analyses, they were performed to see if individual level associations follow the same trend as the occupational

level analysis. Indeed, both the correlations and the ANOVA follow a similar pattern as the occupational level analysis, but the effect sizes are weaker (ranging between $r = -.09$ for self-direction and $r = .07$ for security). This could be expected since the automatability score of individuals is defined at the occupational level, which decreases the variance of automatability independent of variance at the individual level (of values) (see Ostroff, (1993) for further issues with cross-level analysis). These tests are not reported here because they do not produce reliable results. They do, however, confirm that further research at the individual level is needed to fully grasp the complexity of the relationship between automation and values.

Conclusion

The study explores how work-values fit can change due to the technological substitution of work. The study responds to recent calls (Halteh *et al*., 2018; Långstedt and Manninen, 2020) for research on how automation affects person-job fit. The paper argues that the relationship between values and automatability is linked to the different work environments that occupations provide, and the skills required in the occupations. This makes some jobs more appealing to people with certain values and thereby a connection between values and automation emerges. The findings support this argument by indicating that security, conformity, tradition, and power values are positively related to automatability while benevolence, universalism, self-direction, stimulation, and achievement values are negatively related to occupational automatability. Given that the values associated with automatability do not correspond to the skills and abilities assumed to be important in future working life the work-values fit is likely to decrease in the future workplace *if automation progresses as estimated*.

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Appendix A

Share of respondents per country.

	Country	Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Austria	2010	4,5	4,5	4,5
	Belgium	1766	4,0	4,0	8,5
	Switzerland	1525	3,4	3,4	11,9
	Czech Republic	2269	5,1	5,1	17,1
	Germany	2852	6,4	6,4	23,5
	Estonia	2019	4,5	4,5	28,0
	Spain	1958	4,4	4,4	32,4
	Finland	1925	4,3	4,3	36,8
	France	2070	4,7	4,7	41,4
	United Kingdom	1959	4,4	4,4	45,9
	Hungary	1614	3,6	3,6	49,5
	Ireland	2757	6,2	6,2	55,7
	Israel	2557	5,8	5,8	61,5
	Iceland	880	2,0	2,0	63,4
	Italy	2626	5,9	5,9	69,4
	Lithuania	2122	4,8	4,8	74,1
	Netherlands	1681	3,8	3,8	77,9
	Norway	1545	3,5	3,5	81,4
	Poland	1694	3,8	3,8	85,2
	Portugal	1270	2,9	2,9	88,1
	Russian Federation	2430	5,5	5,5	93,6
	Sweden	1551	3,5	3,5	97,1
	Slovenia	1307	2,9	2,9	100,0
	Total	44,387	100,0	100,0	

Appendix B

Excluded occupations from SOC

Low risk	
Teachers and Instructors, All Other	25–3999
Medium risk	
Human Resources, Training, and Labour Relations Specialists, All Other	13–1078
Health Technologists and Technicians, All Other	29–2799
Installation, Maintenance, and Repair Workers, All Other	49–9799
Healthcare Support Workers, All Other	31–9799
Computer Support Specialists	15–1150
High risk	

Appendix C – ANOVA table of value differences in risk groups

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.	*r*
Security	Between Groups	1.264	2	0.632	34.577	0.000	0.59
	Within Groups	2.339	128	0.018			
	Total	3.603	130				
Conformity	Between Groups	0.526	2	0.263	11.560	0.000	0.39
	Within Groups	2.912	128	0.023			
	Total	3.438	130				
Tradition	Between Groups	0.755	2	0.378	12.858	0.000	0.41
	Within Groups	3.759	128	0.029			
	Total	4.515	130				
Benevolence	Between Groups	0.362	2	0.181	11.102	0.000	0.38
	Within Groups	2.085	128	0.016			
	Total	2.446	130				
Universalism	Between Groups	0.368	2	0.184	15.337	0.000	0.44
	Within Groups	1.535	128	0.012			
	Total	1.903	130				
Self-direction	Between Groups	1.444	2	0.722	45.939	0.000	0.65
	Within Groups	2.012	128	0.016			
	Total	3.457	130	0.037			
		0.073	2				
Stimulation	Between Groups	2.746	128	0.021	1.709	0.185	0.16
	Within Groups	2.820	130	0.007			
	Total	0.014	2				
Hedonism	Between Groups	2.348	128	0.018	0.394	0.675	0.08
	Within Groups	2.363	130	0.047			
	Total	0.093	2				
Achievement	Between Groups	3.109	128	0.024	1.921	0.151	0.17
	Within Groups	3.202	130				
	Total	0.269	2	0.134	5.041	0.008	0.27
Power	Between Groups	3.410	128	0.027			
	Within Groups	3.679	130				
	Total	88.094	128	44.047	70.575	0.000	0.72
Highest level of education	Between Groups	79.887	2	0.624			
	Within Groups	167.981	130				
	Total						

Appendix D: Mean occupational values in risk groups

Risk level		Security	Conformity	Tradition	Benevolence	Universalism	Self-direction	Stimulation	Hedonism	Achievement	Power
Low-risk occupations	Mean	0.26	-0.30	-0.06	0.76	0.61	0.56	-0.62	-0.18	-0.42	-0.94
	N	38	38	38	38	38	38	38	38	38	38
	Std. Deviation	0.14	0.13	0.15	0.14	0.12	0.11	0.13	0.10	0.16	0.15
Medium-risk occupations	Mean	0.40	-0.19	0.06	0.74	0.55	0.40	-0.67	-0.16	-0.48	-0.92
	N	40	40	40	40	40	40	40	40	40	40
	Std. Deviation	0.16	0.18	0.21	0.12	0.12	0.16	0.14	0.14	0.18	0.17
High-risk occupations	Mean	0.50	-0.15	0.13	0.64	0.48	0.30	-0.67	-0.19	-0.46	-0.84
	N	53	53	53	53	53	53	53	53	53	53
	Std. Deviation	0.11	0.14	0.15	0.12	0.09	0.11	0.16	0.15	0.13	0.17
	Mean	0.40	-0.20	0.05	0.71	0.54	0.41	-0.66	0.18	-0.45	-0.89
	N	131	131	131	131	131	131	131	131	131	131
	Std. Deviation	0.17	0.16	0.19	0.14	0.12	0.16	0.15	0.13	0.16	0.17